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17 June 1982
SUPERSEDING
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31 August 1972

MILITARY SPECIFICATION

BENDING PRESS, VERTICAL RAM, HYDRAULIC, PIPE, TUBING AND STRUCTURAL SHAPES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of bending press for forming radius bends in pipe, tubing and structural shapes by wrapping the workpiece around the ram or punch die.

1.2 Classification. The presses covered by this specification shall be of the following sizes, as specified (see 6.2.1):

Size 1 - 3 Tons
Size 2 - 5 Tons
Size 3 - 6 Tons
Size 4 - 20 Tons
Size 5 - 30 Tons
Size 6 - 40 Tons

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3441

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and hand books. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Industrial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and the publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402)

(Copies of specifications, standards, handabooks, drawings, and publications required by the manufacture in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 79 - Electrical Standard for Metalworking Machine Tools

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(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210)

(Industry association specifications and standards are generally available for reference from libraries. They are distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 4.3 and 6.2.1), the contractor shall furnish one sample unit for first article inspection and approval.

3.2 Design. The bending press shall be designed to form true radius bends, crush bends, and collapse bends in tubing, pipe, and structural shapes by exerting ram or punch die pressure against opposing rotating wing die pressure to wrap the workpiece around the ram die. The bending press shall be a self-contained, hydraulically operated, vertical machine consisting basically of a base and frame, hydraulic unit, ram, wing die holders, and angle of bend selector. Machines furnished under this specification shall be new and of the manufacturers current design. The design shall be such as to provide convenient, efficient, and safe operation. All parts subject to wear shall be easily accessible for adjustment, repair, or replacement.

3.2.1 Reclaimed materials. The machine manufacture is encouraged to use reclaimed materials to the maximum extent possible without jeopardizing the chemical composition and physical properties of the materials originally slected for the machine construction.

3.2.2 Controls. All operating controls shall be located convenient to the operator's normal work station.

3.2.3 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.3).

3.2.4 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall

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have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.5 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, or filling with solder of paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner that prevents change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.3 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.5 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.6 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.7 Gears. All gears shall be constructed of a material suitable for their application. Gears shall be applied in a manner to preclude either premature damage, wear or failure.

3.4 Performance. The press shall be capable of forming ferrous and non-ferrous tubing, pipe, bars, and structural shapes, with yield point of up to 40,000 psi, to any desired angle from 0 to maximum for the centerline radius as listed in Table I for the size machine in the contract or purchase order. The angle of bend movement shall be controlled by adjustable pins or limit switches and shall be accurate to plus or minus 1/2 degree.

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3.5 Components. The assembled machine shall consist basically of the following components. Any additional components necessary to the operation of the machine shall be furnished to the manufacturer's standard design and specification.

3.5.1 Base and frame. The machine base and frame shall be a casting of iron or steel, or a steel plate weldment, incorporating an oil reservoir and providing for mounting the required components of the machine. Opening shall be provided in the oil reservoir sufficient in size to provide easy cleaning.

3.5.2 Ram. The ram shall be manufactured of settl, machined to mount the bending dies. The ram shall be mounted in sliding ways which are fitted with replaceable wear strips.

3.5.3 Wing die assembly. The wing die assembly shall provide for the "wrapping" or forming of the workpiece around the punch or form die. The wing die holders shall pivot on spindles, provide for the mounting of interchangeagle dies, and for the application of controlled pressure upon the workpiece.

3.5.4 Angle of bend selector. An angle of bend selector consisting of turret mounted pins or limit switches and relays shall be provided. The selector shall provide for automatic cycling of the machine and shall control the angle of bend to plus or minus 1/2 of the selected angle.

3.5.5 Hydraulic system. A system capable of delivering the required hydraulic power shall be provided. Hydraulic pressure shall be provided by one or more electric motor driven pumps. All lines, fittings, valves, and other parts of the hydraulic system, including the reservoir, shall be designed to withstand all pressures, surges, and stresses imposed by 150 percent of operating pressure. The system shall provide pressure relief at 110 percent of operating pressure. The hydraulic system shall automatically unload between machine cycles thereby requiring minimum horsepower to circulate to circulate the fluid. A filtration system designed to prevent particles larger than 40 microns from entering the valves shall be provided. Reservoir openings shall be fitted with gasket sealed covers to prevent the entry of contaminants. Gages shall be installed to show hydraulic fluid level and hydraulic pressure. Unless an oil cooling device or design is included, reservoir capacity shall be adequate to prevent fluid temperature from rising above 150 F.

3.6 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity shall be as specified in table I for the size machine in the contract or order.

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TABLE I

Sizes and Capacities

Hydraulic Power	Size 1 3 Tons	Size 2 5 Tons	Size 3 6 Tons	Size 4 20 Tons	Size 5 30 Tons	Size 6 40 Tons
Maximum Capacity Steel Tubing						
One Tube	1-1/4" OD x .065"	1-1/4" OD x .065"	1-3/8" OD x .065"	2" OD x .083"	3" OD x .125"	3-1/2" OD x .125"
Two Tubes		1" OD x .065"	1" OD x .065"	1" OD x .083"	1-3/4" OD x .125"	2" OD x .125"
Maximum Angle*	140° – 3-1/2 CLR	140° – 3-1/2 CLR	95° – 4-1/2 CLR	165° – 10" CLR	165° – 10" CLR	165° – 10" CLR
Maximum CLR	4-1/2"	4-1/2"	4-1/2"	10"	10"	10"
Minimum CLR	2D	2D	2D	5"	4"	4-1/2"

* Increasing the centerline radius above those listed will reduce the maximum angle proportionately.

3.7 Electrical equipment. Unless otherwise specified (see 6.2.1), all electrical components including motors, motor starters, relays, switches, enclosures, and wiring shall conform to and shall be located in accordance with NFPA-79. The machine shall be completely wired through the controls and ready for connection to the applicable power source.

3.7.1 Motors. Motors shall be the squirrel cage, induction type, with sealed for life type anti-friction bearings. Unless otherwise specified (see 6.2.1), motors shall be dripproof dual voltage 230/460 volt, three phase, 60 Hz, AC initially wired for operation on 230 volts.

3.8 Equipment.

3.8.1 Standard equipment. The machine shall be furnished with all standard equipment normally furnished by the manufacturer.

3.8.2 Optional equipment. When required (see 6.2.1), optional equipment shall be furnished as specified.

3.9 Nameplate. Unless otherwise specifed (see 6.2.1), a corrosion resistant metal nameplate shall be securely affixed to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the nameplate shall contain the model number of the basic standard machine.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Power input characteristics and ratings

Date of manufacture

Contract Number of Order Number

National Stock Number or Plant Equipment Code

3.10 Workmanship. Workmanship of the machine and accessories shall be commensurate with the requirements specified herein and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his

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own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First Article inspection. When First Article inspection is required, it shall be applied to the First Article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), First Article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the First Article to pass the examination of one or more First Article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1. Failure of the item to pass the examination or test shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined for design, construction, materials, components, electrical equipment, and workmanship to determine compliance with the requirements of this specification.

4.6 Tests.

4.6.1 Operational test. The machine shall be operated at no load for a minimum of 30 minutes during which time the controls, wing dies, hydraulic system, ram, and all other operating components of the machine shall be acutated and observed to determine proper operation.

4.6.2 Performance test. The machine shall be set up to bend workpieces of maximum capacity and to the maximum degree of bend for the machine being tested. The machine shall then be operated to bend six (6) workpieces, at least two feet long, to the same angle of bend without further adjustment. The six (6) workpieces shall be examined to determine if the degree of bend is within 1/2 degree of the set up bend and is free from wrinkles or distortion.

4.7 Packaging inspection. Packaging shall be inspected to determine comliance with the requirements of Section 5.

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5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. Machines produced to this specification are intended for use in bending pipe, tubing, and structural shaspes in production shops where tolerances of plus or minus 1/2 degree are required.

6.2 Ordering data.

6.2.1 Acquistion documents. Acquistion documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size of machine (see 1.2 and 3.6).

c. Specify additional safety and health requirements (see 3.2.3 and 6.3).

d. Painting, if different (see 3.3.4).

e. Electrical equipment, if different (see 3.7).

f. Voltage of initial wiring, if different (see 3.7.1).

g. Motor enclosure, if different (see 3.7.1).

h. Optionaly equipment, if required (see 3,8,2).

i. Nameplate, if different (see 3.9).

j. First article inspection, if different (see 4.3).

k. Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instruction for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorportated into the contract.

6.3 Safety and health requirements. Paragraph 3.2.3 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation

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levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the requesting activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, as in most cases, the guard configuration is dependent on the size and configuration of workpieces. The above, and any other additional safety and health requirements, should be specified in detail under 6.2.1(c).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - SH
Air Force - 99

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User Activities:
Army - MU, SM
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Preparing Activity:
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DOCUMENT IDENTIFIER AND TITLE
MIL-B-80160B Bending Press, Vertical Ram, Hydraulic Pipe, Tubing and Structural Shapes

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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